

Q. Name the following scientists:

- A) Father of Genetics
- B) Discovery of Chromosomes
- C) Proposed Chromosomal theory of Inheritance
- D) Experimental evidence of Chromosomal theory of inheritance
- E) Discovery of Nuclein
- F) Proposed Transformation
- G) Confirmation DNA as genetic material
- H) Semiconservative model of replication
- I) Isolation of DNA Polymerase I
- J) Invention of PCR
- K) Chain termination reaction

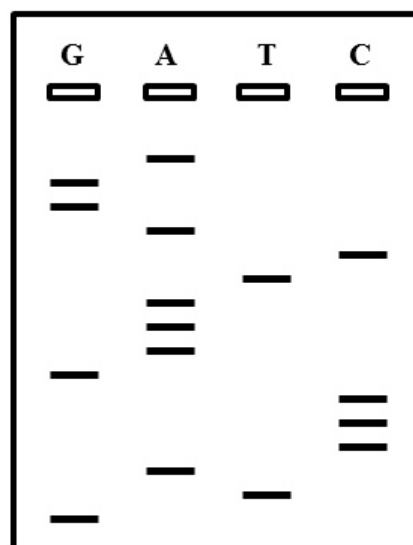
Q What is the difference between a deoxynucleotide and a dideoxynucleotide?

Q. When setting up a Sanger sequencing reaction, each reaction should include template DNA, nucleotides, dideoxynucleotides, buffer, DNA polymerase, and ____.

Q. What happens in automated Sanger DNA sequencing reaction if you forget to add:

- a) all fluorescent ddNTPs?
- b) fluorescent ddGTP?
- c) all dNTPs?
- d) dGTP?

Q. Based on the following gel picture find out the sequence of DNA sample?



Q. In DNA sequencing by the Sanger (dideoxy) method:

- A) radioactive dideoxy ATP is included in each of four reaction mixtures before enzymatic synthesis of complementary strands.**
- B) specific enzymes are used to cut the newly synthesized DNA into small pieces, which are then separated by electrophoresis.**
- C) the dideoxynucleotides must be present at high levels to obtain long stretches of DNA sequence.**
- D) the role of the dideoxy CTP is to occasionally terminate enzymatic synthesis of DNA where Gs occur in the template strands.**
- E) the template DNA strand is radioactive.**

Q. Describe briefly what is meant by saying that two DNA strands are complementary.

Q. What is the approximate length of a DNA molecule (in the B form) containing 10,000 base pairs?

Q. Triple-helical DNA structures can result from Hoogsteen (non Watson-Crick) interactions. These interactions are primarily:

- A) covalent bonds involving deoxyribose.**
- B) covalent bonds involving the bases.**
- C) hydrogen bonds involving deoxyribose.**
- D) hydrogen bonds involving the bases.**
- E) hydrophobic interactions involving the bases.**

Q. B-form DNA in vivo is a _____-handed helix, _____ Å in diameter, with a rise of _____ Å per base pair.

- A) left; 20; 3.9**
- B) right; 18; 3.4**
- C) right; 18; 3.6**
- D) right; 20; 3.4**
- E) right; 23; 2.6**

Q. The double helix of DNA in the B-form is stabilized by:

- A) covalent bonds between the 3' end of one strand and the 5' end of the other.**
- B) hydrogen bonding between the phosphate groups of two side-by-side strands.**
- C) hydrogen bonds between the riboses of each strand.**

- D) nonspecific base-stacking interaction between two adjacent bases in the same strand.
- E) ribose interactions with the planar base pairs.

Q. The DNA oligonucleotide abbreviated pATCGAC:

- A) has seven phosphate groups.
- B) has a hydroxyl at its 3' end.
- C) has a phosphate on its 3' end.
- D) has an A at its 3' end.
- E) violates Chargaff's rules.

Q. The PCR reaction mixture does *not* include:

- A) all four deoxynucleoside triphosphates.
- B) DNA containing the sequence to be amplified.
- C) DNA ligase.
- D) heat-stable DNA polymerase.
- E) oligonucleotide primer(s).

Q. What happens in PCR if you use DNA polymerase derived from *E. coli* instead of from a thermostable source?

Q. The Meselson-Stahl experiment established that:

- A) DNA polymerase has a crucial role in DNA synthesis.
- B) DNA synthesis in *E. coli* proceeds by a conservative mechanism.
- C) DNA synthesis in *E. coli* proceeds by a semiconservative mechanism.
- D) DNA synthesis requires dATP, dCTP, dGTP, and dTTP.
- E) newly synthesized DNA in *E. coli* has a different base composition than the preexisting DNA.

Q. When a DNA molecule is described as replicating bidirectionally, that means that it has two:

- A) chains.
- B) independently replicating segment.
- C) origins.
- D) replication forks.
- E) termination points.

Q. The DNA below is replicated from left to right. Label the templates for leading strand and lagging strand synthesis.

(5')ACTTCGGATCGTTAAGGCCGCTTTCTGT(3')

(3')TGAAGCCTAGCAATTCCGGCGAAAGACA(5')

Q. The *E. coli* chromosome contains 4.64×10^6 bp.

(a) How many turns of the double helix must be unwound during replication of the *E. coli* chromosome?

(b) How long would it take to replicate the *E. coli* chromosome at 37 °C

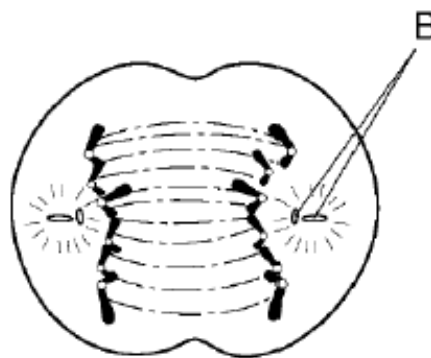
if two replication forks proceeded from the origin? Assume replication occurs at a rate of 1,000 bp/s.

Under some conditions *E. coli* cells can divide every 20 min. How might this be possible?

(c) In the replication of the *E. coli* chromosome, about how many Okazaki fragments would be formed? What factors guarantee that the numerous Okazaki fragments are assembled in the correct order in the new DNA?

Q.

The cell in the diagram below illustrates a stage of mitotic cell division.



Letter *B* indicates the

- A. paired chromosomes
- B. centrioles
- C. cell plate
- D. endoplasmic reticulum

Q.

Which mitotic event in the chart occurs after the other three events have taken place?

<i>A</i>	Appearance of spindle fibers
<i>B</i>	Separation of chromatids by the action of spindle fibers
<i>C</i>	Disintegration of the nuclear membrane
<i>D</i>	Replication of chromosomes

A. A B. B C. C D. D

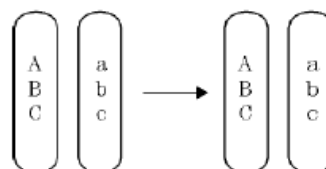
Q.

The process of mitosis usually involves

- A. chromosome duplication and synapsis
- B. DNA replication and separation of chromatids
- C. tetrad formation and fertilization
- D. reduction in chromosome number and formation of cell plate

Q.

Occurs in meiosis- not mitosis
Which process, occurring during synapsis, would result in the chromosomal changes illustrated in the diagram.



- A. replication
- B. crossing-over
- C. independent assortment
- D. segregation

Q.

Suppose that a cell is unable to replicate its DNA. Which of the following would NOT be present in this cell?

- a. sister chromatids
- b. centromere
- c. DNA
- d. genes
- e. homologous chromosomes

Q.

During meiosis, chromosome number

- a. Doubles from the original cell, then divides in four for each daughter cell.
- b. Divides in four for each daughter cell.
- c. Doubles, then divides in half.
- d. Divides in half.
- e. Doubles.

During meiosis I,

- a. Homologous chromosomes separate.
- b. Each sister chromatid becomes a chromosome.
- c. Sister chromatids separate.
- d. 4 daughter cells are formed from a single original cell.
- e. Two of the above.

The main difference between meiosis and mitosis is that

- a. DNA replicates during mitosis, but does not during meiosis.
- b. During mitosis, sister chromatids separate; they do not during meiosis.
- c. Mitosis makes genetically identical copies; meiosis does not.
- d. Mitosis increases chromosome number in each cell, while meiosis decreases it.
- e. None of the above.